



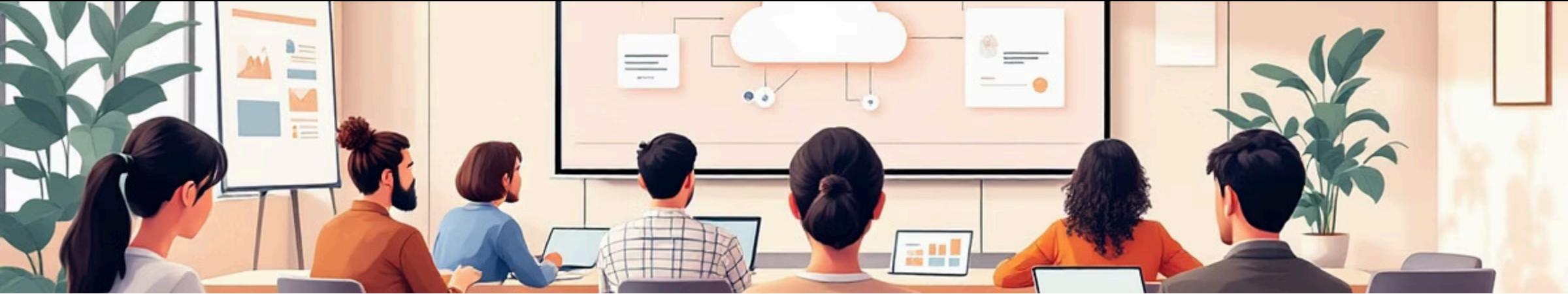
AWS 3-Tier Architecture Design & Deployment

A comprehensive internship project demonstrating scalable cloud infrastructure deployment using Amazon Web Services

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Roll No. : 5649

S. M. Joshi College, Pune



COMPANY PROFILE

Exlearn Technologies

Practical technology training for learners and professionals.

Focused on building job-ready skills in cloud computing, DevOps, and project-based learning.

IT Training & Development

Cloud and DevOps training for strong technical foundations.

Hands-On Projects

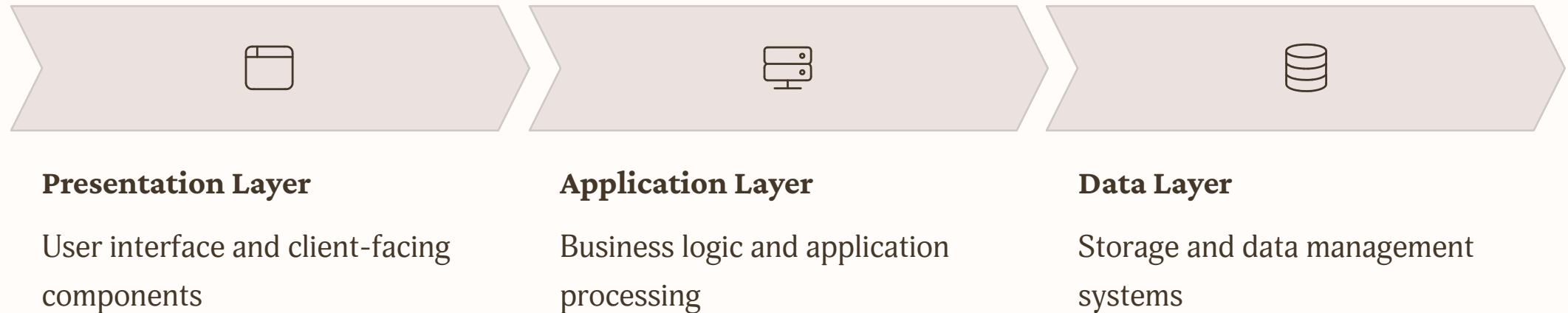
Industry-style projects for practical application.

Industry Experience

Workplace exposure and career preparation.

Project Overview

This project demonstrates the deployment of a production-ready **3-tier web application** on AWS infrastructure, focusing on enterprise-grade scalability and reliability.



The architecture emphasises scalability, high availability, and comprehensive monitoring capabilities.

Project Objectives

01

Scalable Cloud Architecture

Design a resilient, horizontally scalable infrastructure using AWS best practices

02

AWS Service Integration

Deploy application components using appropriate AWS services and configurations

03

Auto Scaling & Load Balancing

Implement dynamic scaling groups and elastic load balancing for traffic management

04

Centralised Logging

Enable comprehensive monitoring and log aggregation using CloudWatch

05

CDN Performance

Improve content delivery speed and reduce latency using CloudFront CDN

Technical Stack

Application Components

- **Frontend:** React.js for responsive user interface
- **Backend:** Node.js for server-side logic
- **Database:** MySQL on RDS for data persistence

AWS Services Utilised

- EC2 (Elastic Compute Cloud)
- RDS (Relational Database Service)
- CloudFront (Content Delivery Network)
- Route53 (DNS Management)
- Auto Scaling Group
- CloudWatch (Monitoring)

Key Advantages



High Availability

Redundant components ensure continuous service uptime and minimal downtime



Auto Scaling

Dynamic resource allocation based on traffic demand for optimal performance



Centralised Monitoring

Comprehensive logging and real-time metrics using CloudWatch



Fault Tolerance

Automatic failover and recovery mechanisms protect against infrastructure failures



Secure Architecture

Multi-layer security with encrypted data transmission and access controls



CDN Acceleration

Global content delivery network reduces latency and improves user experience

Practical Applications

This 3-tier architecture pattern is applicable to various enterprise scenarios requiring scalable web infrastructure:

E-Commerce Platforms

Online stores with high traffic volumes and transaction processing requirements

Blogging Systems

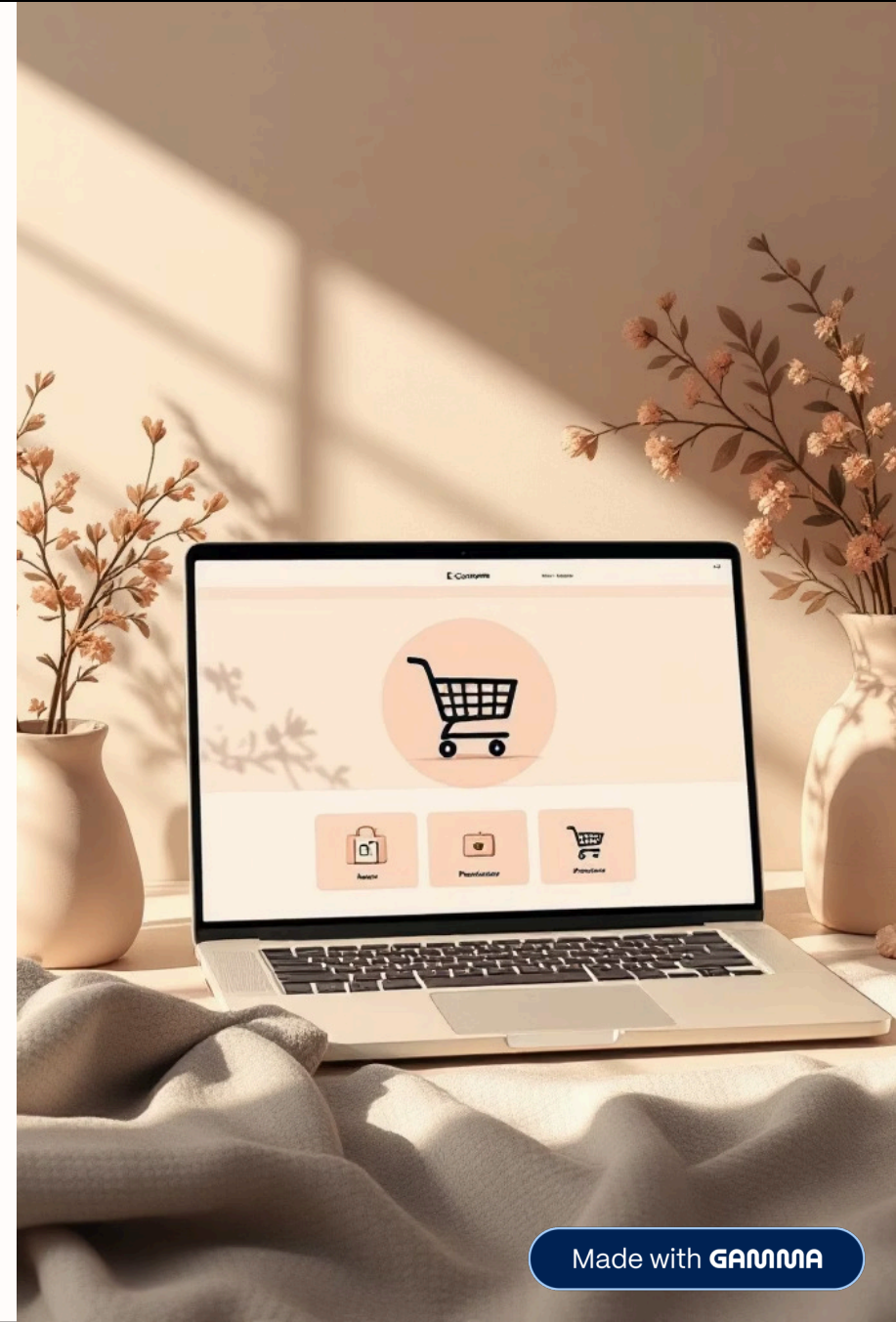
Content management platforms with dynamic content generation

Enterprise Applications

Business systems requiring robust security and reliability

SaaS Platforms

Multi-tenant software-as-a-service applications serving multiple customers



Entity Relationship Diagram

Database Structure

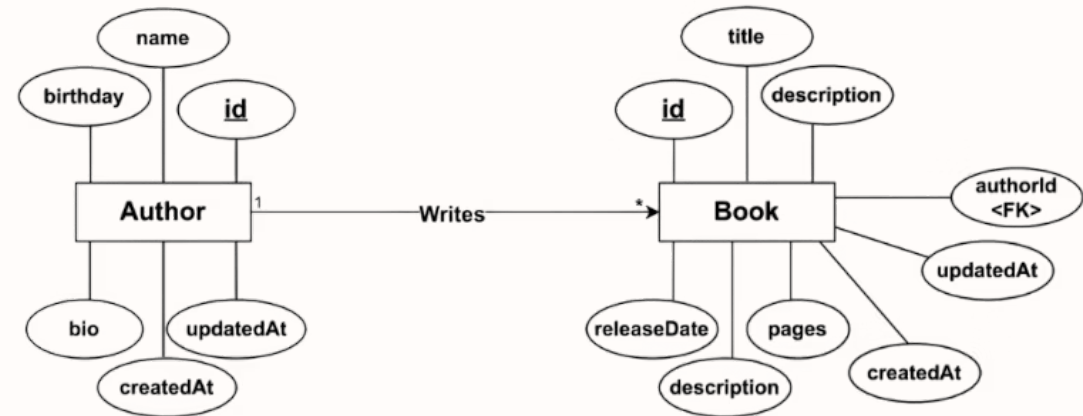
The ER diagram illustrates the relational database design, showing entities and their relationships for data modelling purposes.

Entities Defined

- **Author:** Stores author information
- **Book:** Contains book details

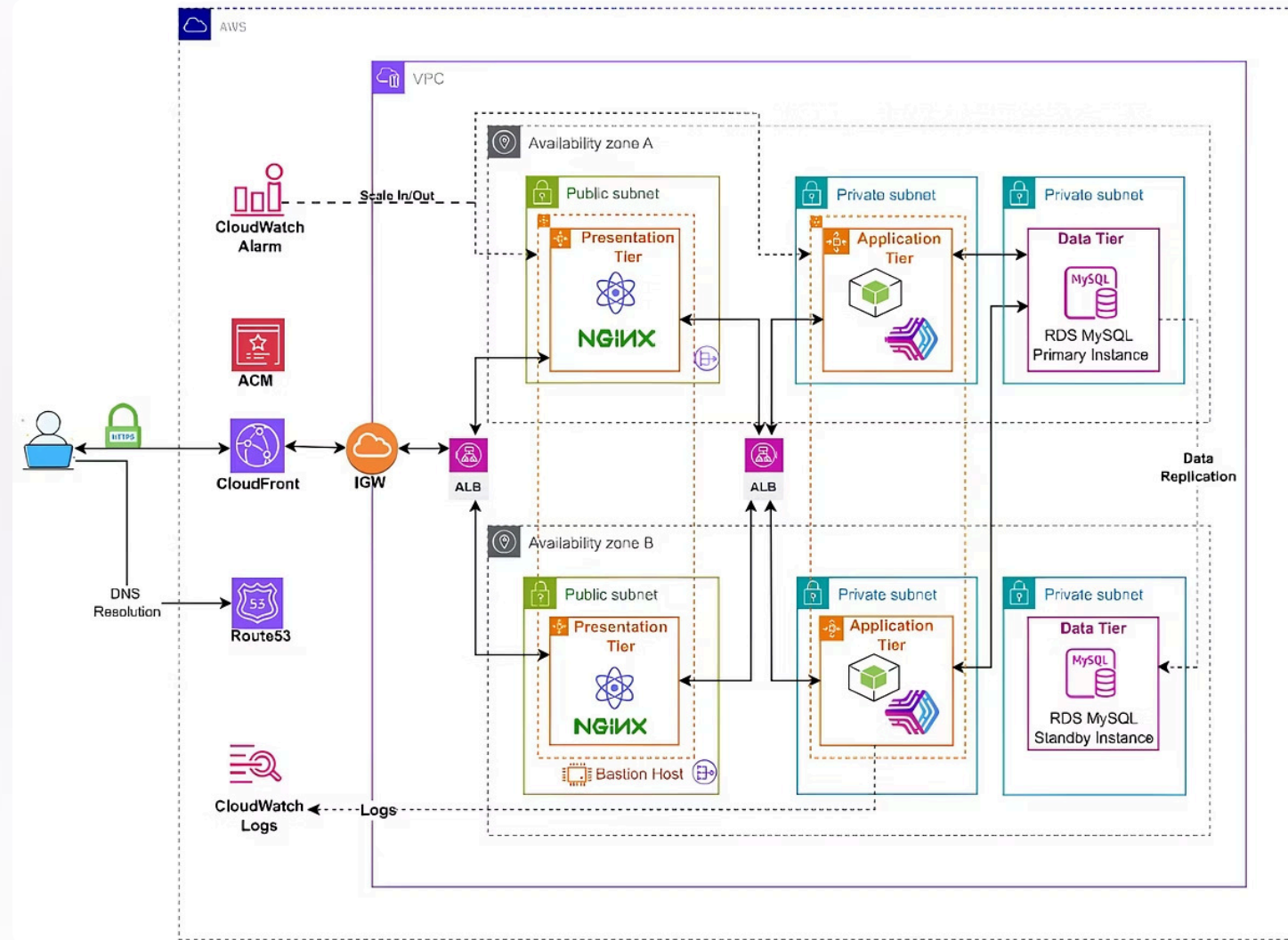
Relationship Type

One-to-Many relationship where **one Author** can write **multiple Books**.



System Architecture

This diagram shows a three-tier architecture with presentation, application, and data layers working together to deliver the system. It supports scalability and clear separation of concerns by keeping each layer focused on its own role.

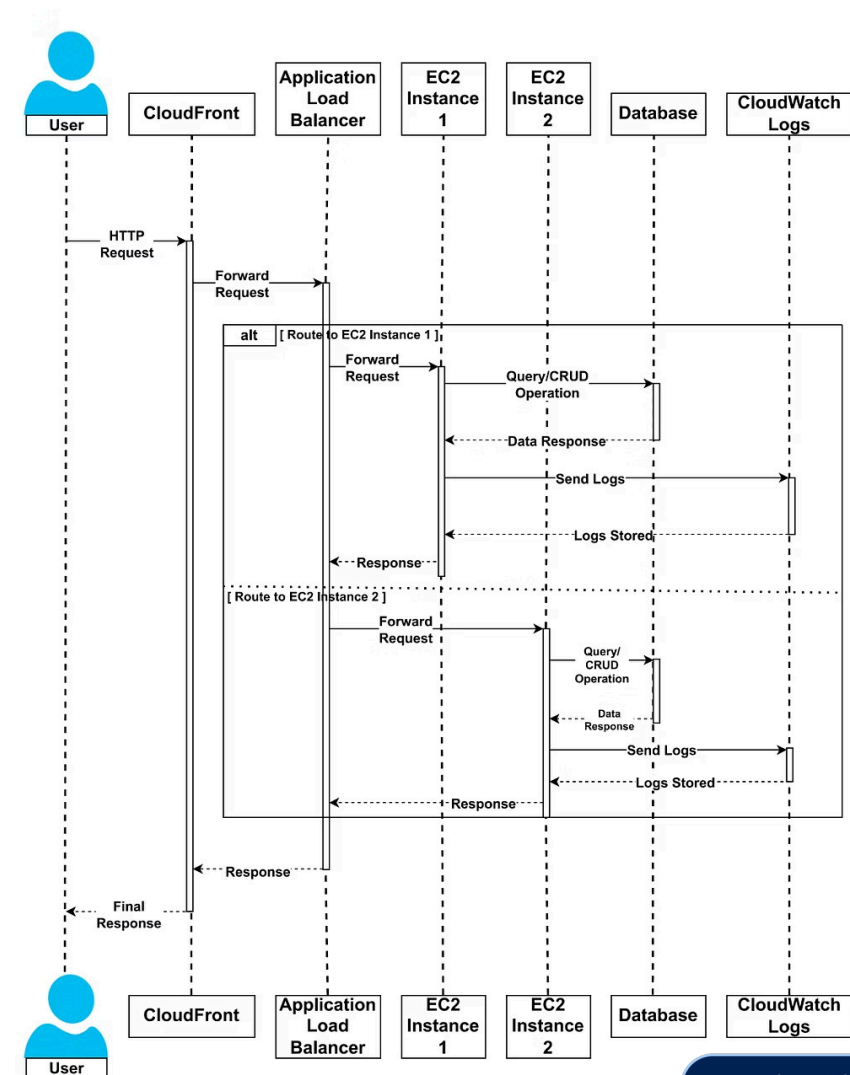


Sequence Diagram

This diagram shows the interaction flow between the User, CloudFront, Application Load Balancer, EC2 instances, Database, and CloudWatch Logs.

Requests flow from the user through each layer of the system, where they are processed and logged as needed.

Responses then return back through the same path to complete the interaction.

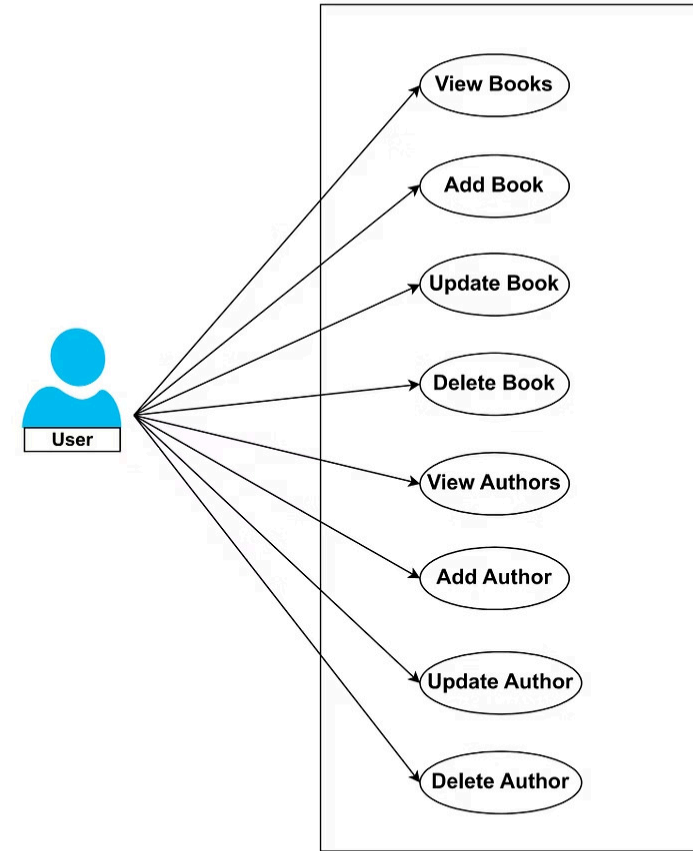


Use Case Diagram

This diagram shows all the interactions between the user and the system.

The main use cases include View Books, Add Book, Update Book, Delete Book, View Authors, Add Author, Update Author, and Delete Author.

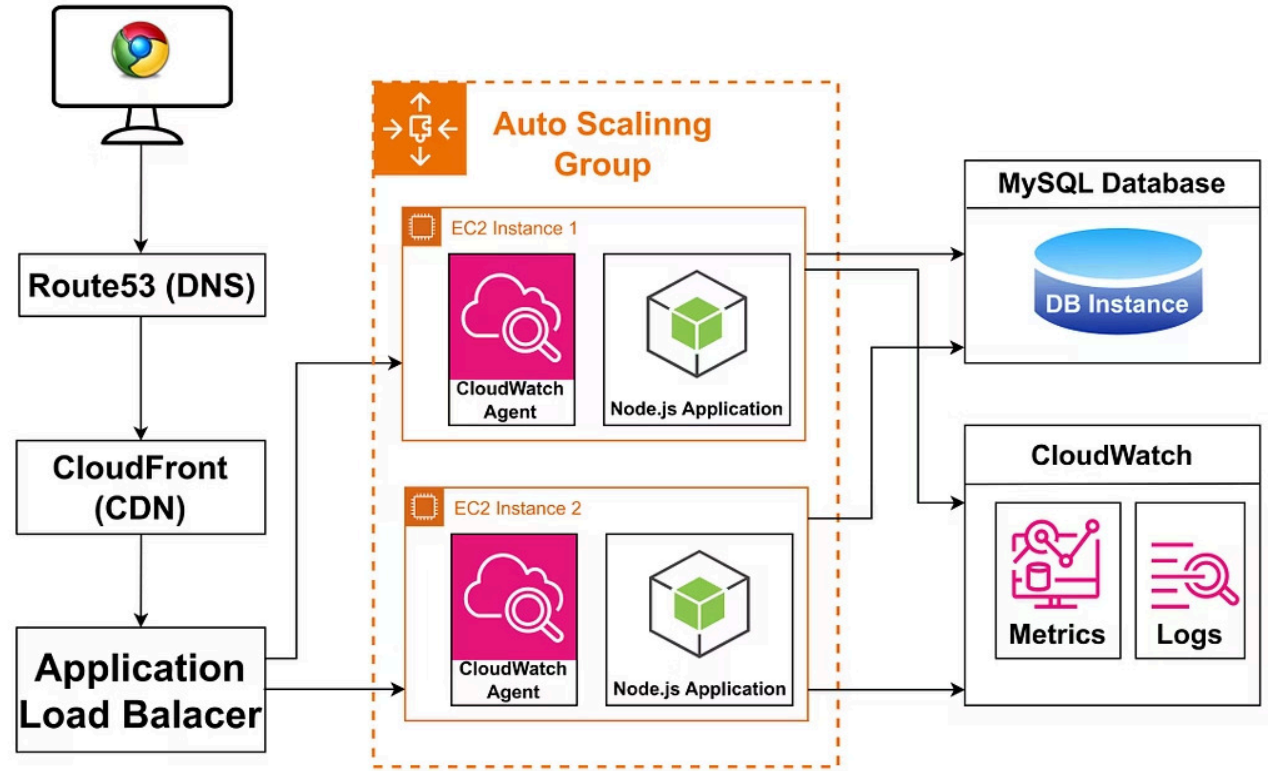
Together, these represent the core functionalities available to users in the application.



Deployment Diagram

This diagram shows the physical deployment of the application across AWS infrastructure, including Route53 DNS, CloudFront CDN, an Application Load Balancer, an Auto Scaling Group with EC2 instances, a MySQL database, and CloudWatch monitoring.

Together, these components work to ensure scalability, availability, and performance across the system.



Implementation Details (Screenshots)

1 Domain and DNS Configuration

2 VPC and Network Setup

3 Bastion Host and Database Setup

4 Presentation Tier Deployment & Testing

5 Application Tier Deployment & Testing

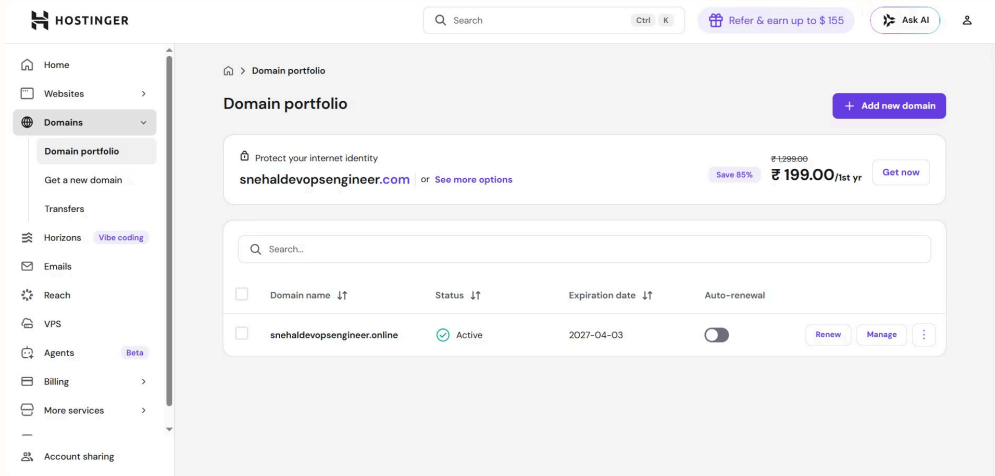
6 Application User Interface

7 Monitoring and Logging

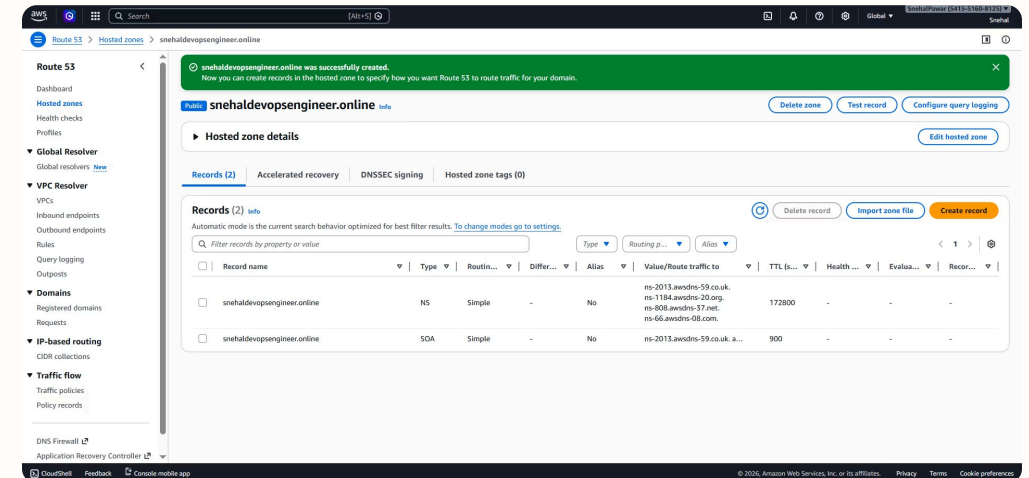
8 Content Delivery and DNS Mapping

9 Final Output (Domain & HTTPS Access)

1. Domain and DNS Configuration

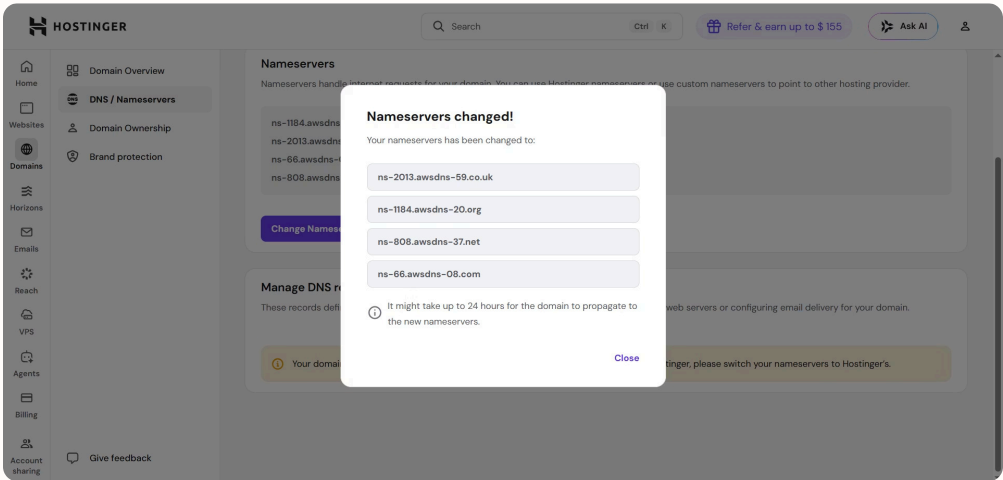


1.1 Domain Registration

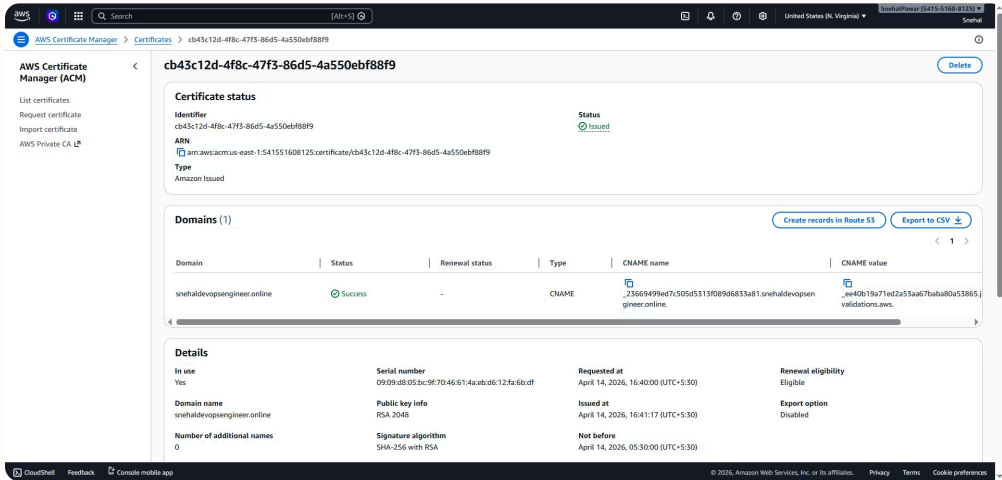


1.2 Hosted Zone Creation

1. Domain and DNS Configuration



1.3 Domain Name Server Update



1.4 SSL Certificate Configuration (ACM)

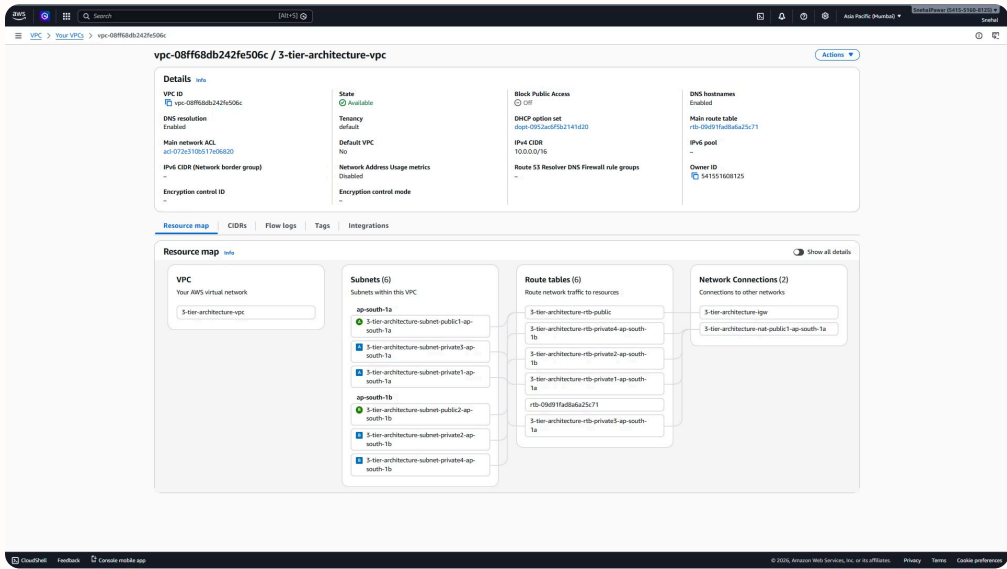
1. Domain and DNS Configuration

The screenshot shows the AWS Certificate Manager console. The breadcrumb trail is: AWS Certificate Manager > Certificates > c237aa5f-c8ef-4a63-985c-8af0c5e0da00 > Create DNS records in Amazon Route 53. The main heading is 'Create DNS records in Amazon Route 53 (1/1)'. Below this is a search bar with the text 'Search domains' and a '1 match' indicator. There are three filter buttons: 'Validation status = Pending validation', 'Validation status = Failed', and 'Is domain in Route 53? = Yes'. A 'Clear filters' button is also present. Below the filters is a table with the following columns: Domain, Validation status, and Is domain in Route 53? The table contains one row with the domain 'snehaldevopsengineer.online', a 'Pending validation' status, and a 'Yes' value for 'Is domain in Route 53?'. At the bottom right of the table are 'Cancel' and 'Create records' buttons. The footer of the console shows 'CloudShell', 'Feedback', 'Console mobile app', and copyright information for Amazon Web Services, Inc. or its affiliates.

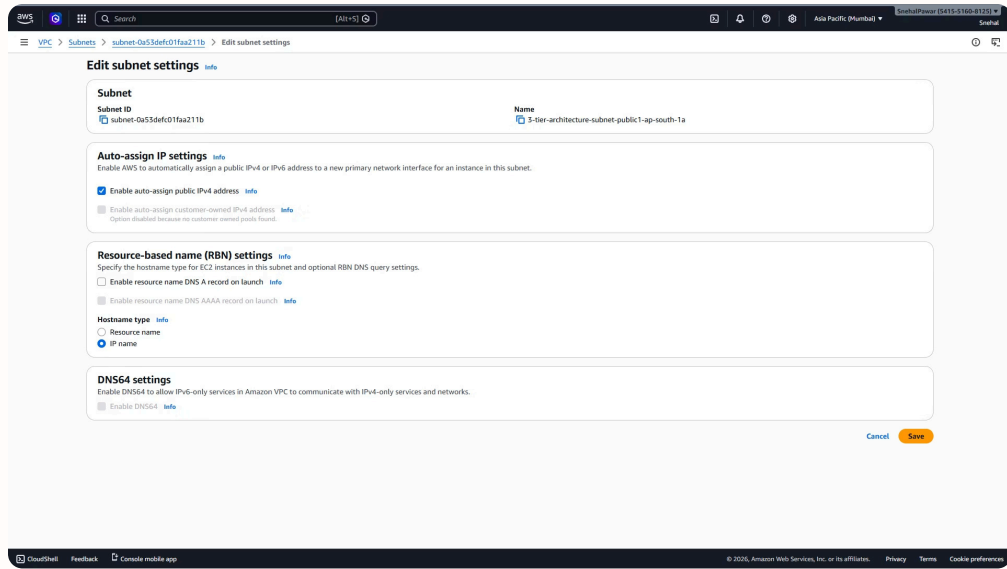
Domain	Validation status	Is domain in Route 53?
snehaldevopsengineer.online	Pending validation	Yes

1.5 DNS Record Configuration in Route53

2. VPC and Network Setup

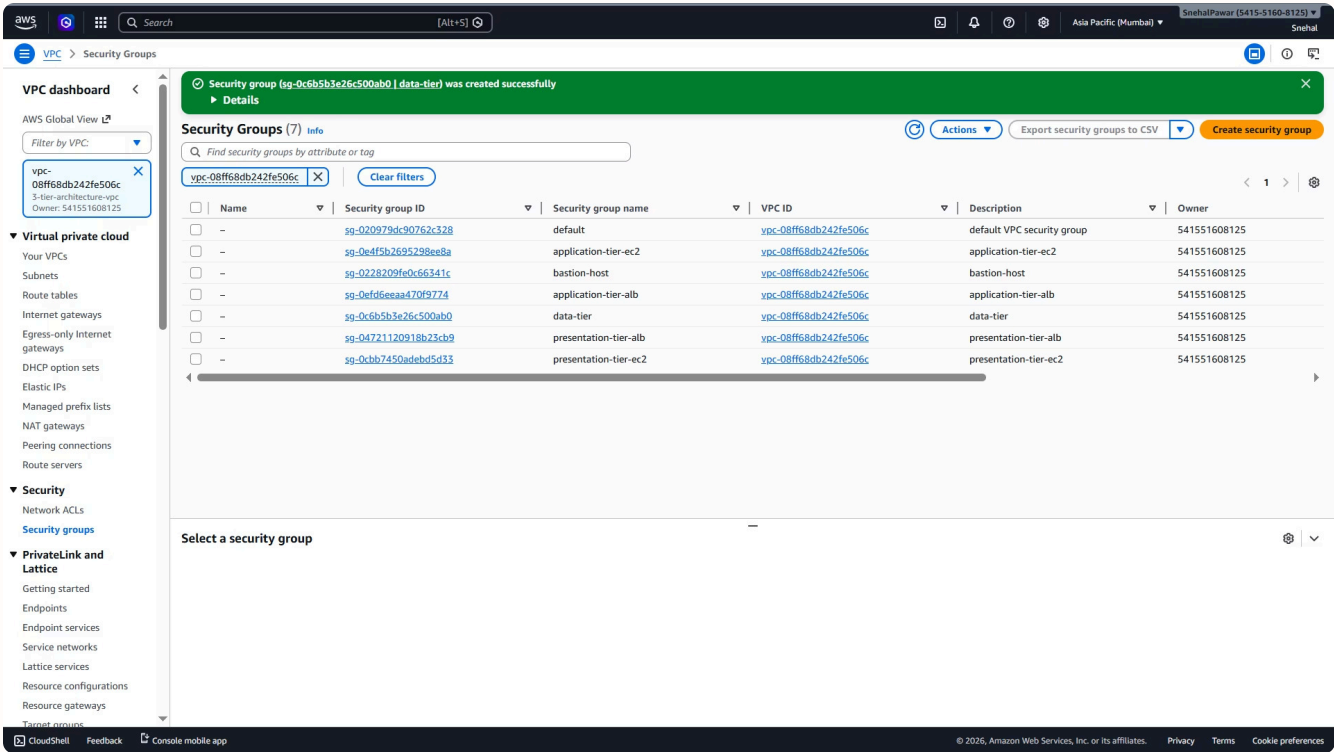


2.1 Virtual Private Cloud (VPC) Creation



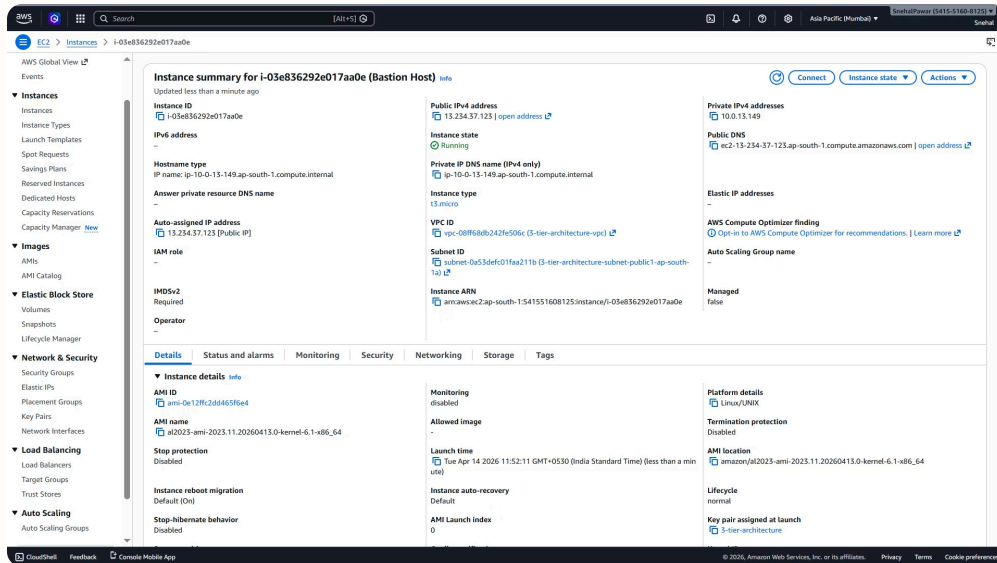
2.2 Public Subnet Configuration with Auto-Assign IP

2. VPC and Network Setup

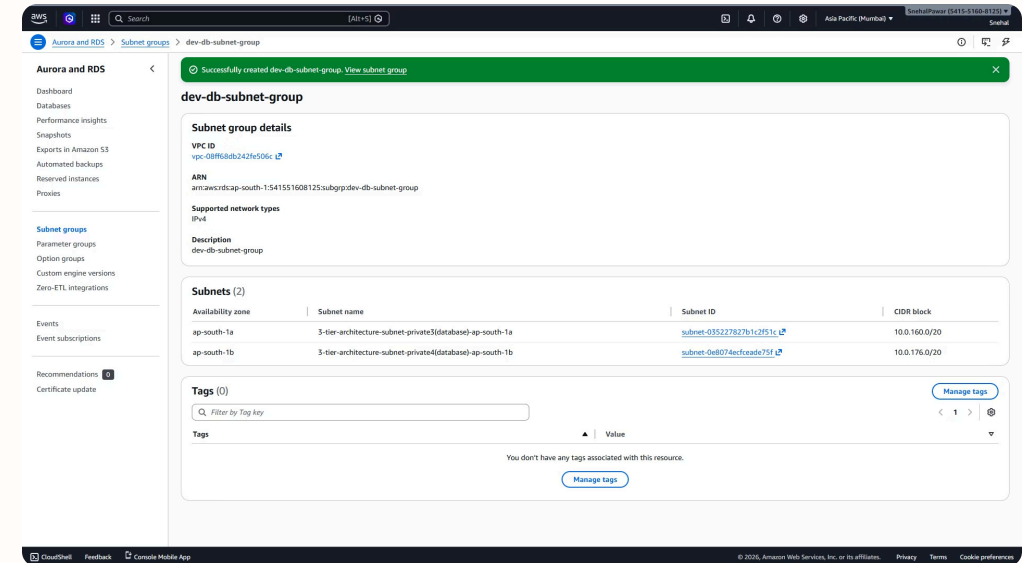


2.3 Security Group Configuration

3. Bastion Host and Database Setup

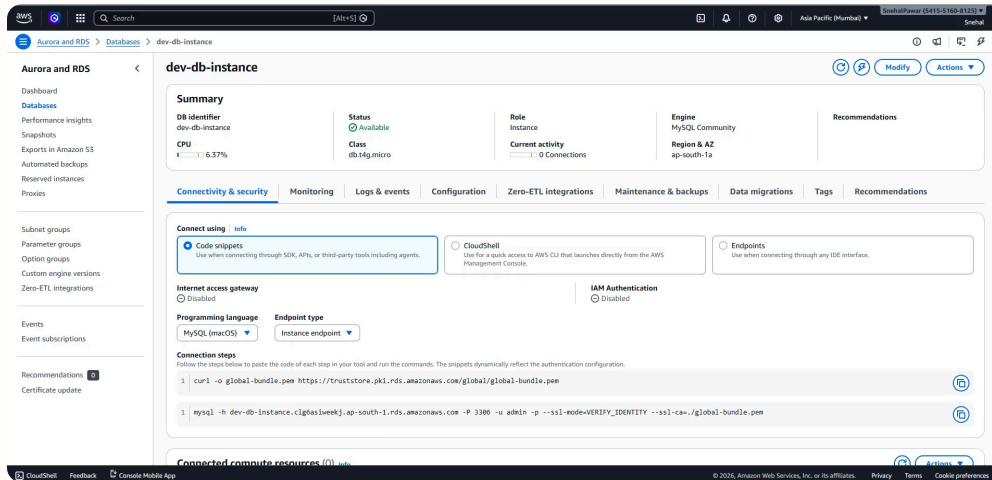


3.1 Bastion Host EC2 Instance Setup

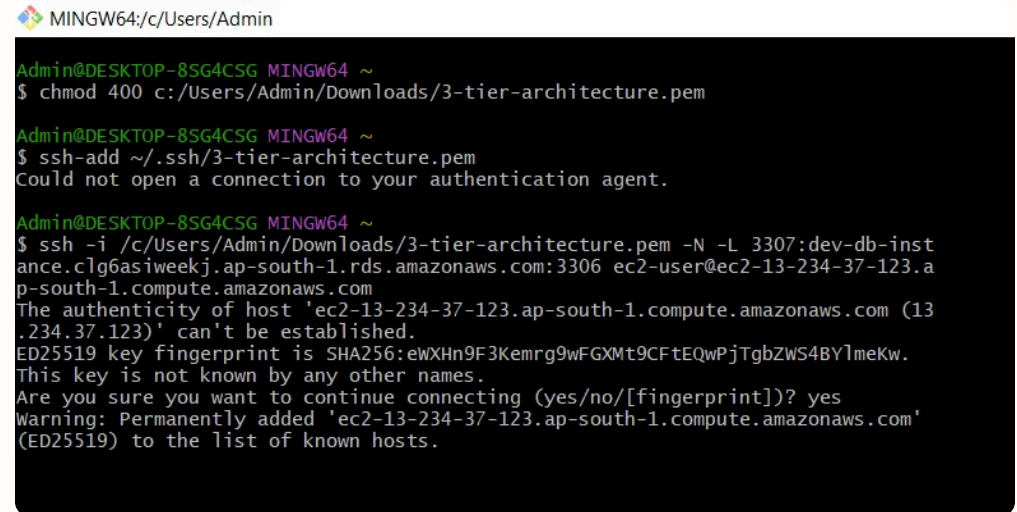


3.2 DB Subnet Group Configuration

3. Bastion Host and Database Setup

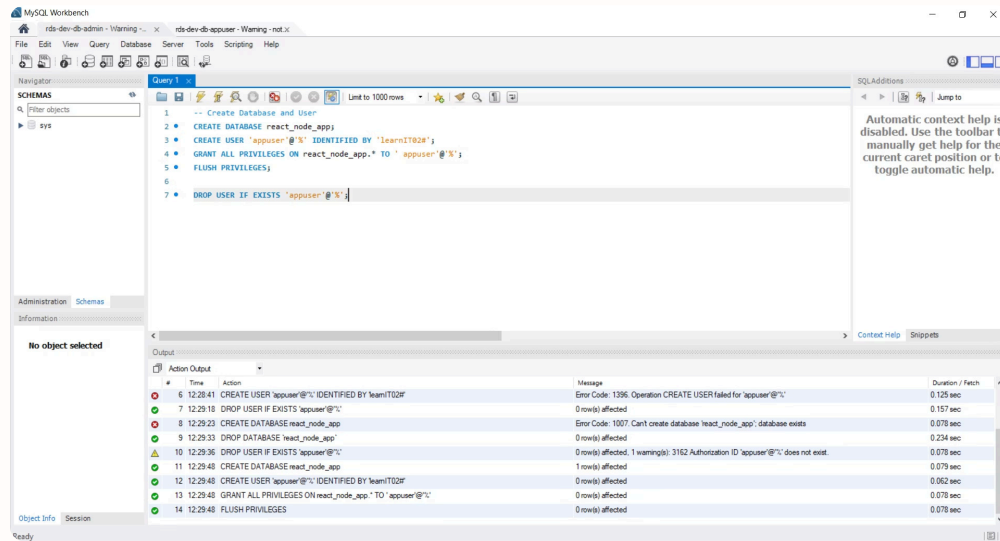


3.3 Amazon RDS Instance Creation

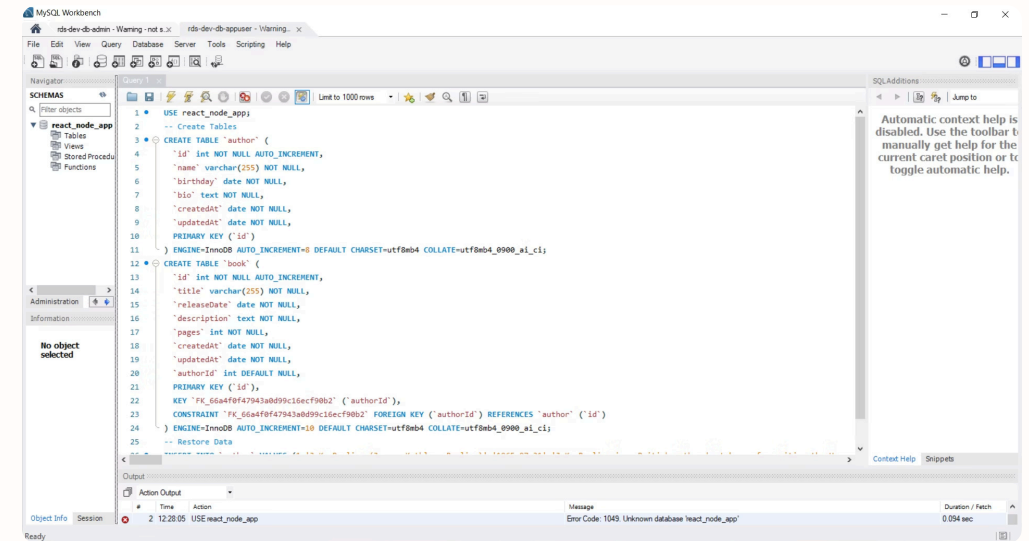


3.4 Secure RDS Access via SSH Tunnelling

3. Bastion Host and Database Setup

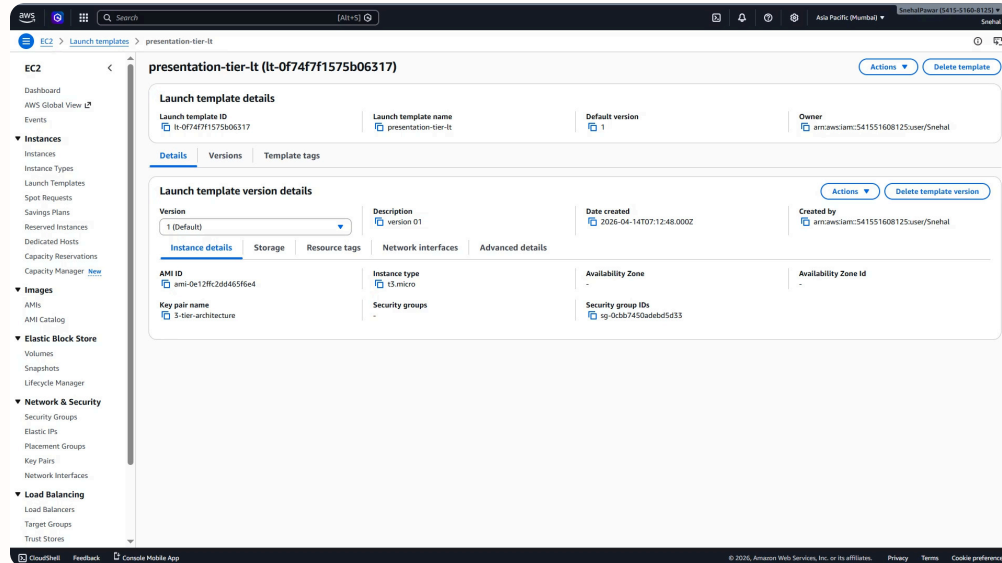


3.5 Database Setup (Admin User)

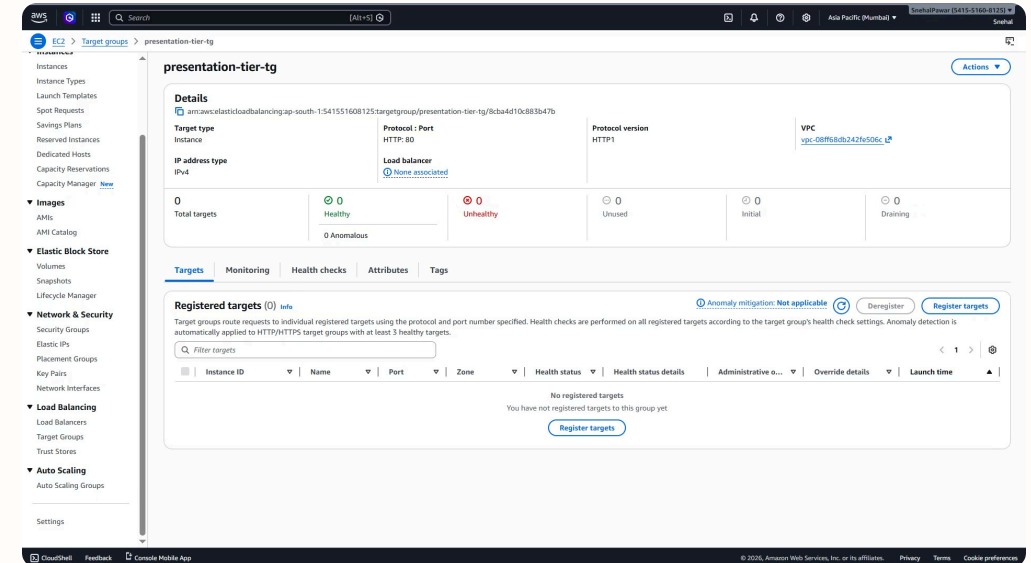


3.6 Database Setup (App User)

4. Presentation Tier Deployment and Testing

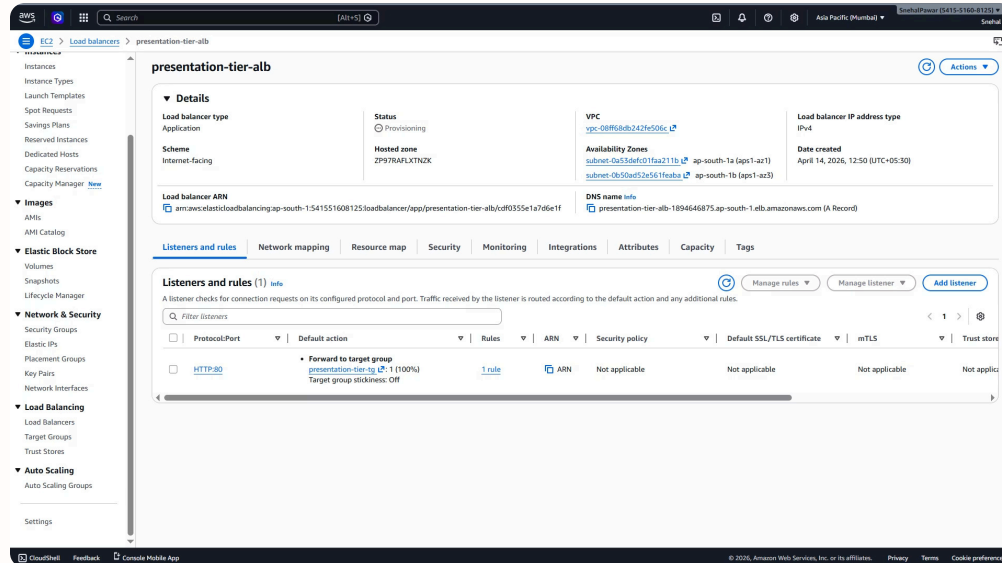


4.1 Launch Template Configuration - Presentation Tier

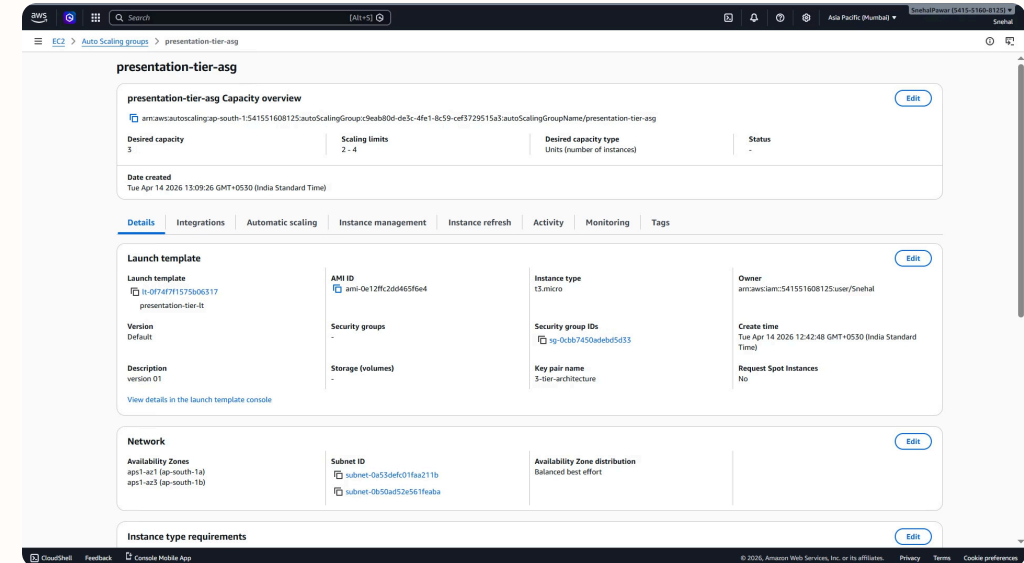


4.2 Target Group Setup - Presentation Tier

4. Presentation Tier Deployment and Testing

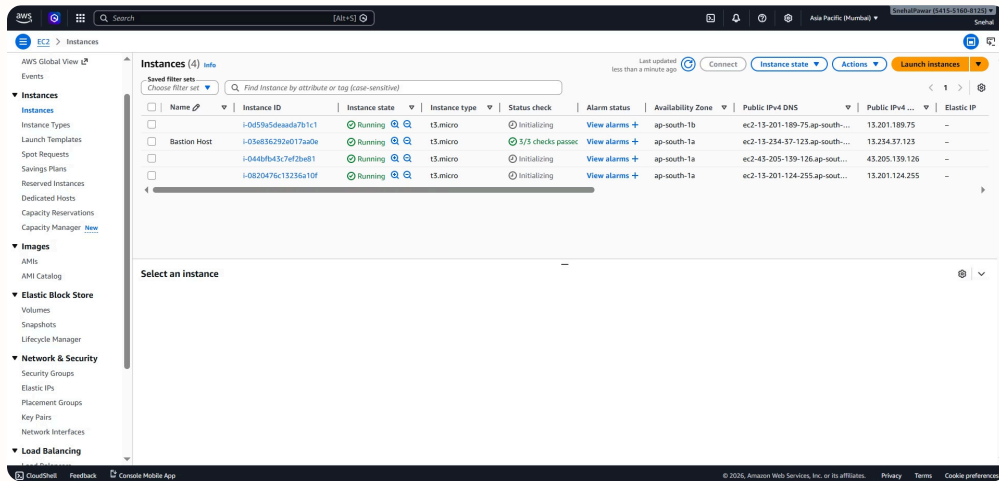


4.3 Application Load Balancer - Presentation Tier

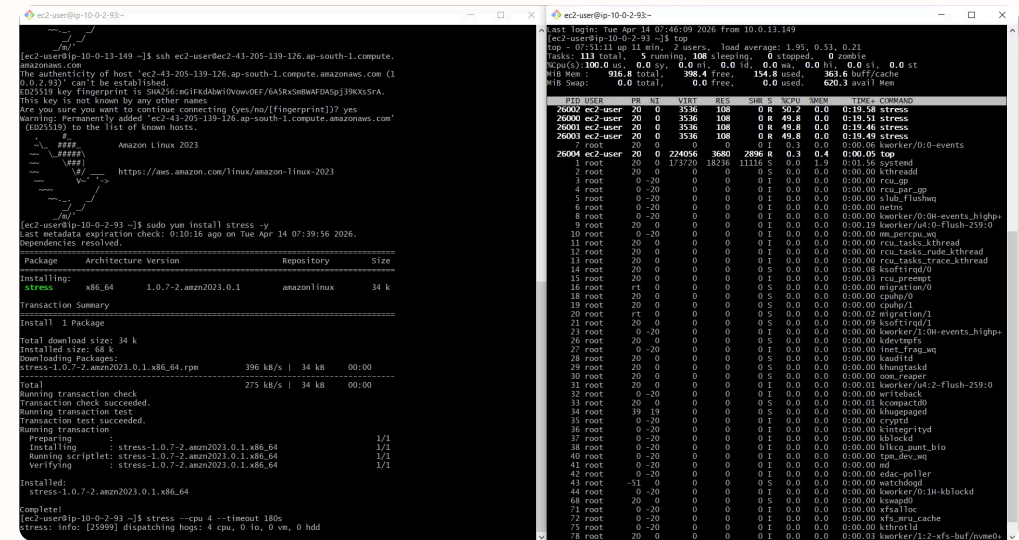


4.4 Auto Scaling Group - Presentation Tier

4. Presentation Tier Scaling and Testing

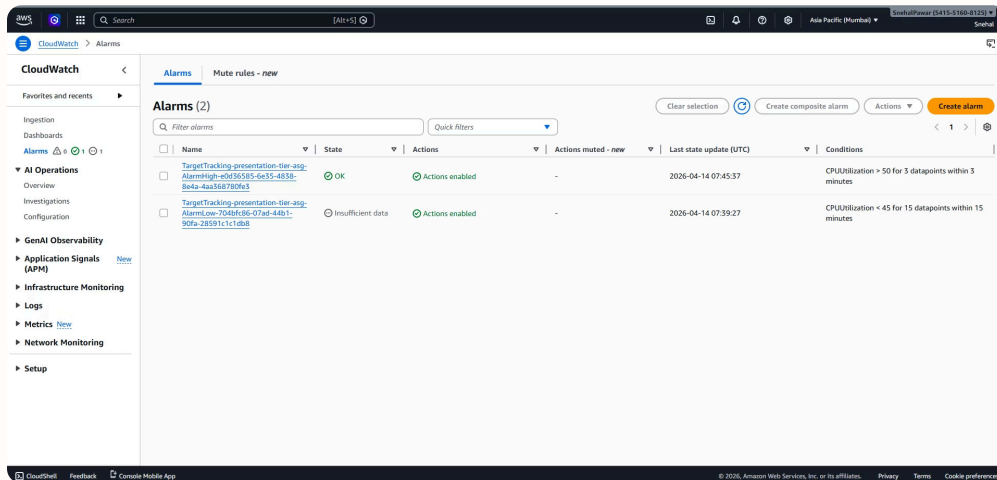


4.5 EC2 Instance Created by Auto Scaling Group

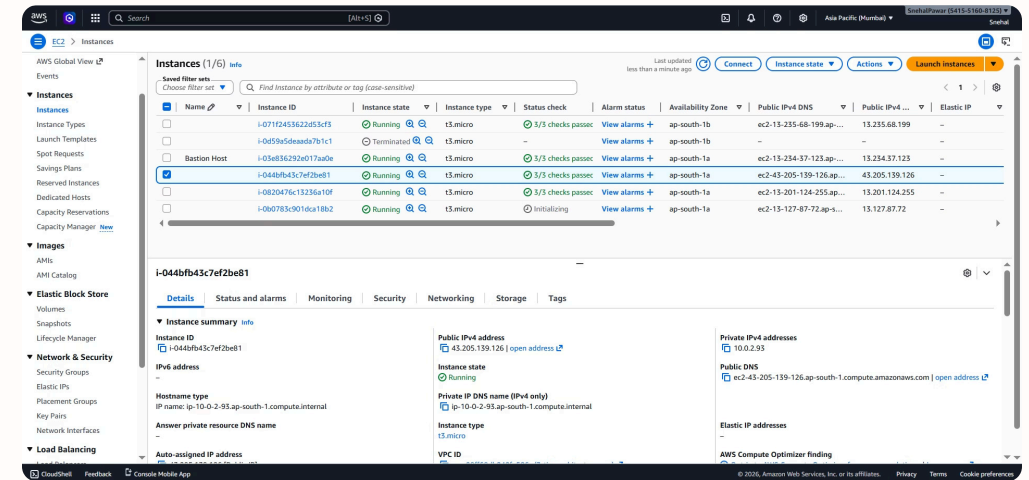


4.6 Load Generation

4. Presentation Tier Scaling and Testing

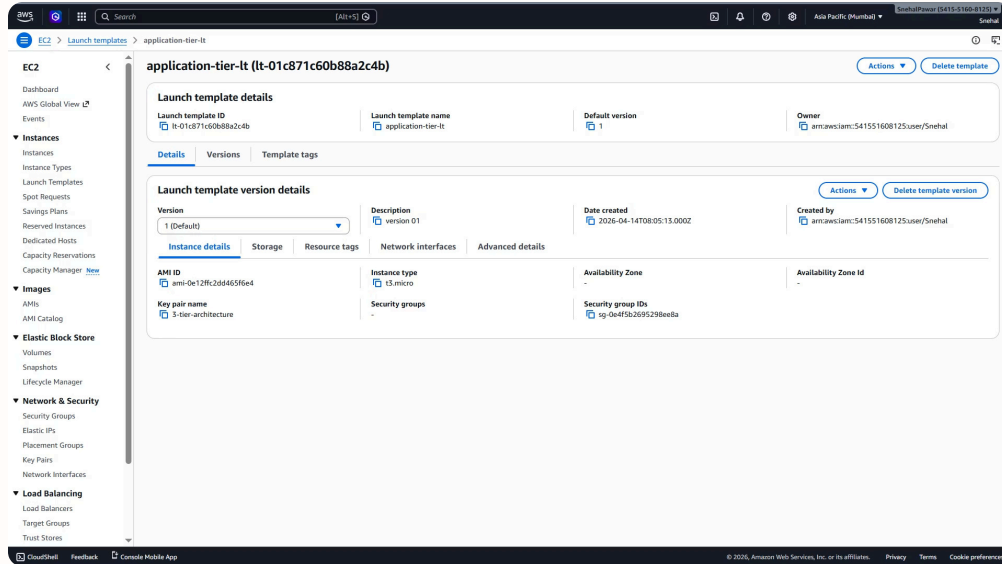


4.7 CloudWatch Alarm Triggered

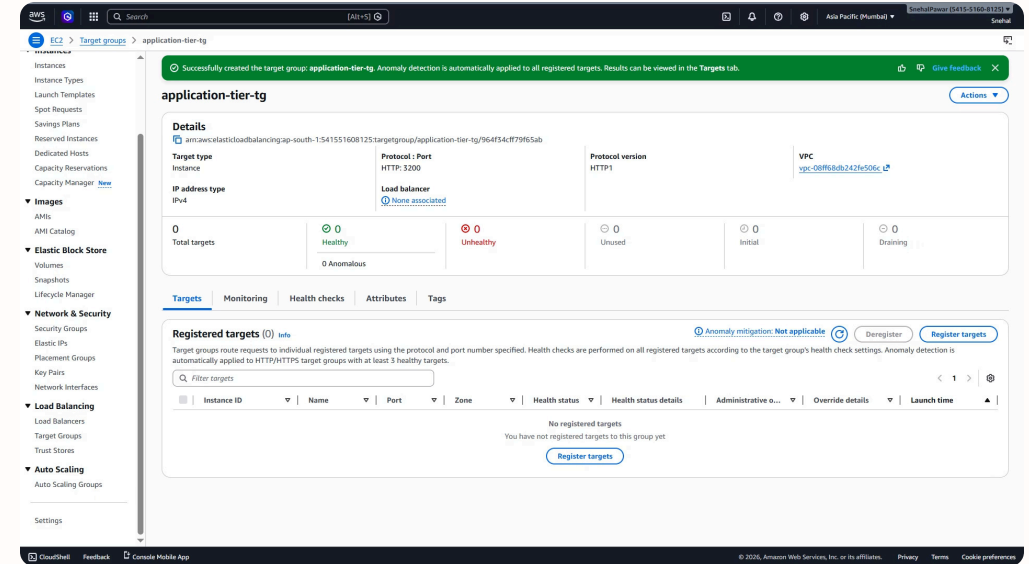


4.8 Auto Scaling Activity

5. Application Tier Scaling and Testing

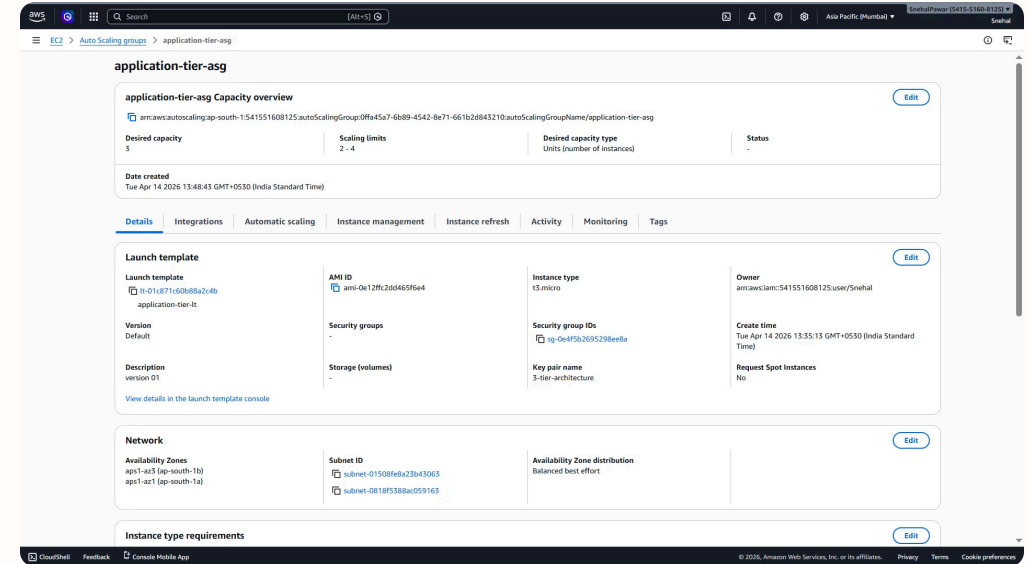
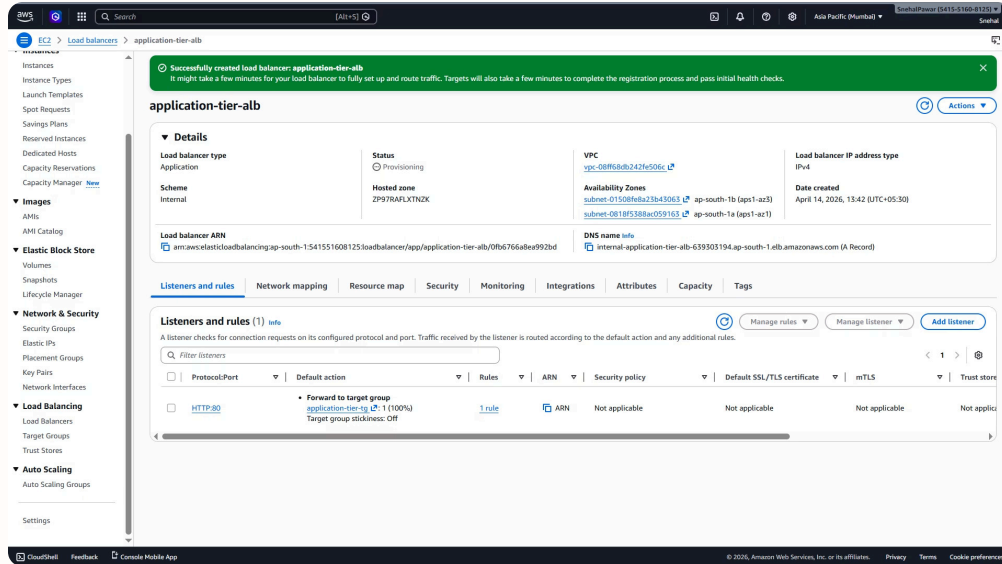


5.1 Launch Template Creation - Application Tier



5.2 Target Group - Application Tier

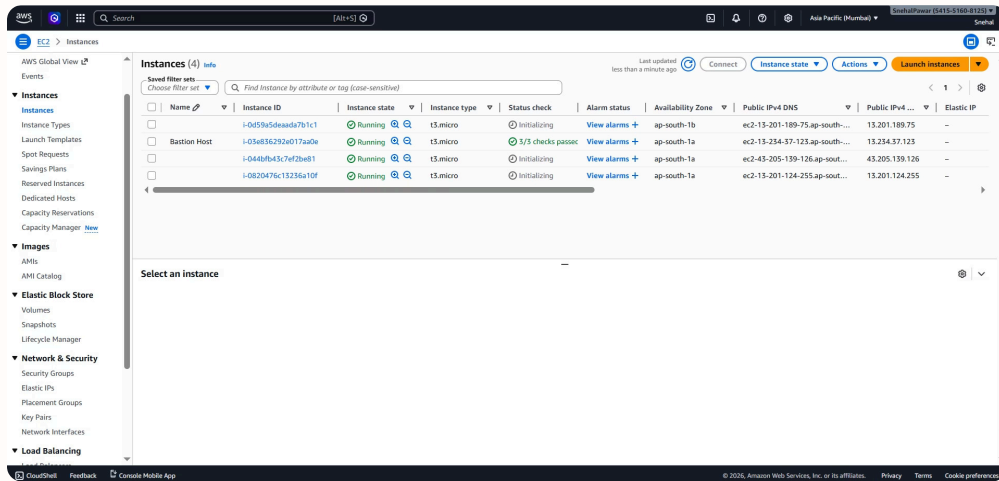
5. Application Tier Deployment and Testing



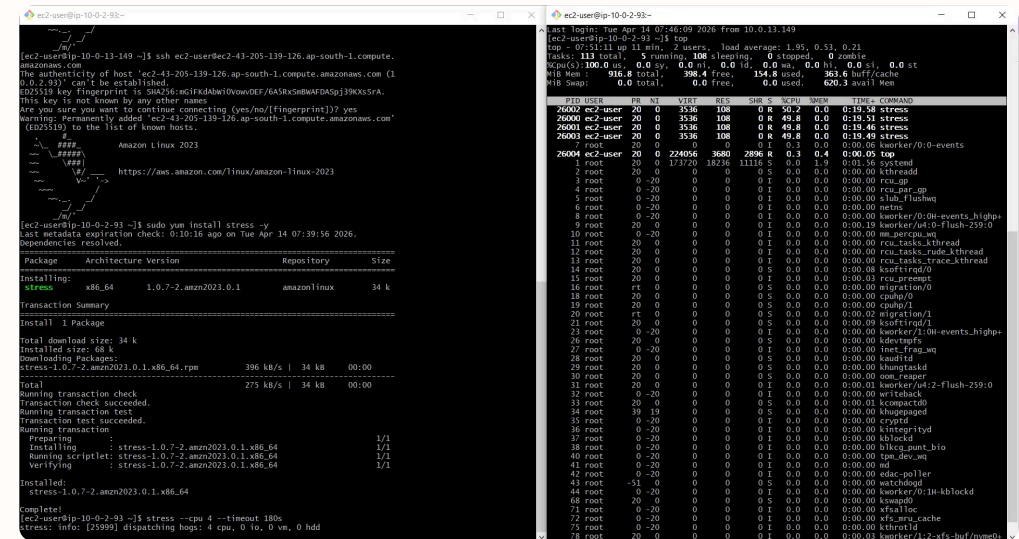
5.3 Application Load Balancer - Application Tier

5.4 Auto Scaling Group - Application Tier

5. Application Tier Scaling and Testing

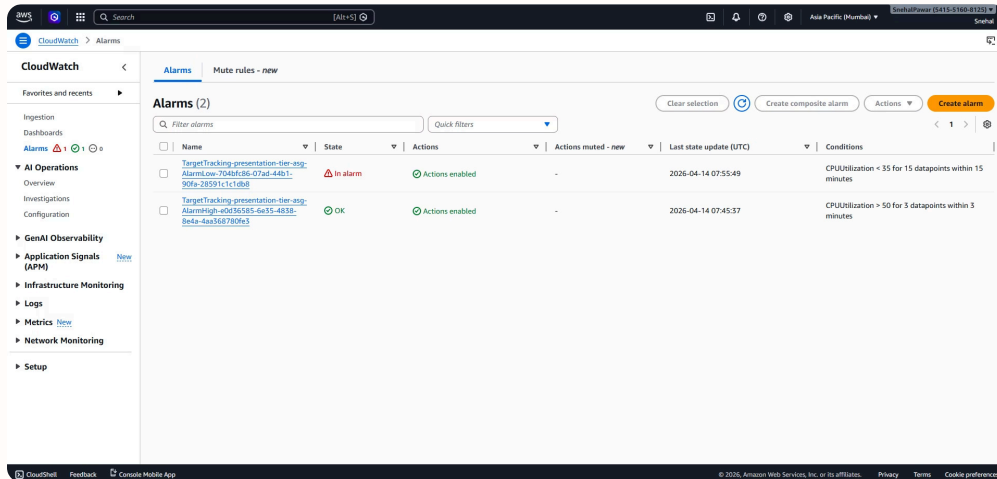


5.5 EC2 Instance Created by Auto Scaling Group

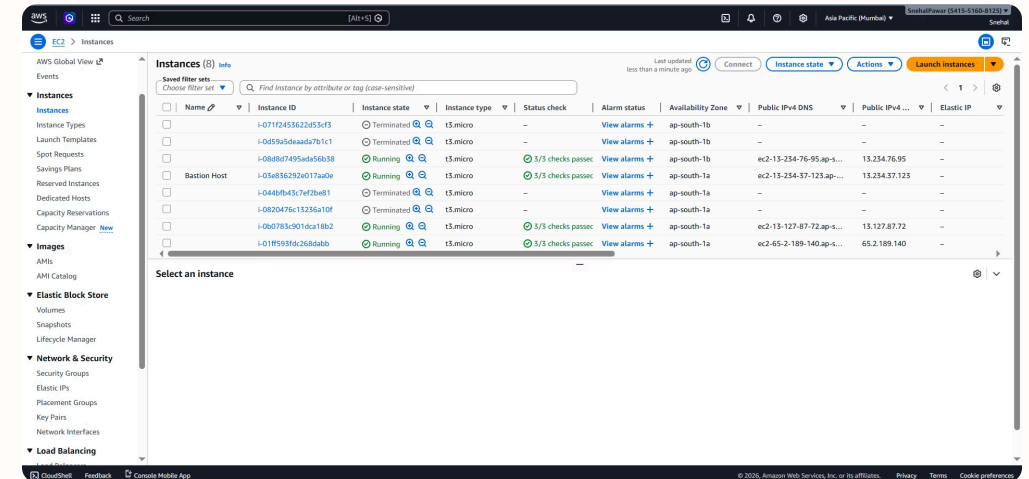


5.6 Load Generation

5. Application Tier Scaling and Testing

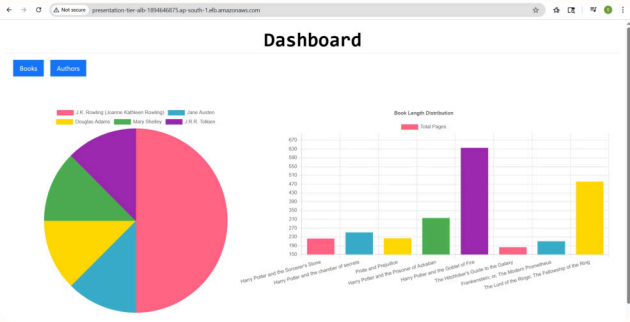


5.7 CloudWatch Alarm Triggered



5.8 Auto Scaling Activity

6. Application User Interface



6.1 Application Home Page

MANAGE AUTHORS

+ Add Author

ID	Author	Birthday	Description	Created Date	Updated Date	Actions
1	J.K. Rowling (J.K. Rowling)	1965-07-31	J.K. Rowling is a British author best known for writing the Harry Potter fantasy series. The series has won multiple awards and sold over 500 million copies, becoming the best-selling book series in history. Rowling has also written other novels, including The Casual Vacancy and the Cormoran Strike crime series under the pen name Robert Galbraith.	2024-05-29	2024-05-29	[Edit] [Delete]
3	Jane Austen	1775-12-16	Jane Austen was an English novelist known for her wit, social commentary, and romantic stories. Her six major novels, which explore themes of love, marriage, and money, have earned her a place as one of the greatest writers in the English language.	2024-05-29	2024-05-29	[Edit] [Delete]
4	Harper Lee	1926-04-28	Harper Lee was an American novelist best known for her Pulitzer Prize-winning novel To Kill a Mockingbird. The novel explores themes of racial injustice and the importance of compassion. Lee published a sequel, Go Set a Watchman, in 2015.	2024-05-29	2024-05-29	[Edit] [Delete]
5	J.R.R. Tolkien	1894-01-29	J.R.R. Tolkien was a British philologist and writer best known for his fantasy novels The Hobbit and The Lord of the Rings. Tolkien's works have had a profound influence on the fantasy genre and popular culture.	2024-05-29	2024-05-29	[Edit] [Delete]
6	Mary Shelley	1818-03-01	Mary Shelley was a British novelist, playwright, and short story writer, the daughter of Mary Wollstonecraft Godwin and the wife of poet Percy Bysshe Shelley. Frankenstein, or The Modern Prometheus (1818) is her most famous work.	2024-05-29	2024-05-29	[Edit] [Delete]
7	Douglas Adams	1929-10-11	Douglas Adams was an English science fiction writer, satirist, humorist, dramatist, screenwriter, and occasional actor. He is best known for the Hitchhiker's Guide to the Galaxy comedy series, which inspired a radio comedy, several books, three shows, comic books, a 1981 TV series, a 1984 video game, a 2005 feature film, and a 2008 musical film.	2024-05-29	2024-05-29	[Edit] [Delete]

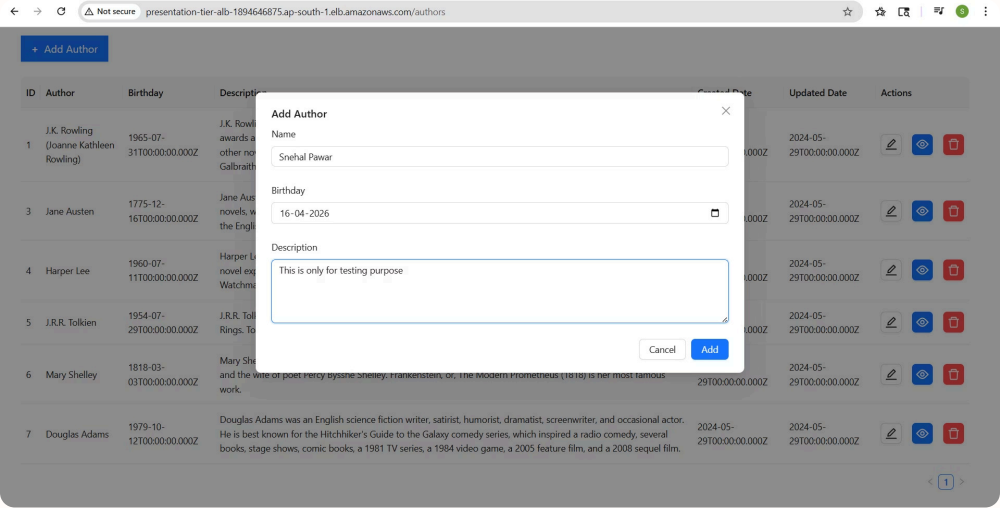
6.2 Authors Management Page

MANAGE BOOKS

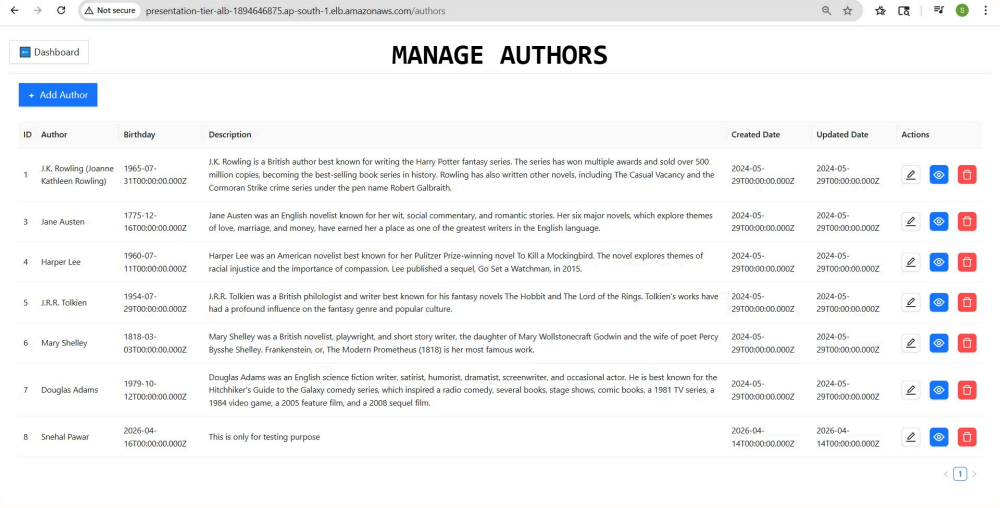
ID	Title	Description	Release Date	Author	Created Date	Updated Date	Actions
1	Harry Potter and the Sorcerer's Stone	On his birthday, Harry Potter discovers that he is the son of two witch-blood parents, then returns to his adopted magical parents. He must attend a famous school of magic and secrecy where he establishes a friendship with two young men who will become his companions on his adventures. During his first year at Hogwarts, he discovers that a malicious and powerful wizard named Voldemort is in search of a philosopher's stone that grants the life of the immortal.	1997-07-21	J.K. Rowling (J.K. Rowling)	2024-05-29	2024-05-29	[Edit] [Delete]
3	Harry Potter and the Chamber of Secrets	Harry Potter and the mysterious investigations of a mysterious threat to their Hogwarts' existence, a menacing threat that hides within the walls.	1998-07-11	J.K. Rowling (J.K. Rowling)	2024-05-29	2024-05-29	[Edit] [Delete]
4	Pride and Prejudice	An English novel of manners by Jane Austen, first published in 1813. The story centres on the relationships among the Bennet sisters, in particular Elizabeth Bennet, the middle daughter, and the wealthy Mr. Darcy, Austen satirizes the social class of the English gentry through witty and ironic narrative voice.	1813-01-28	Jane Austen	2024-05-29	2024-05-29	[Edit] [Delete]
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08	J.K. Rowling (J.K. Rowling)	2024-05-29	2024-05-29	[Edit] [Delete]
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament in which three schools of wizardry will compete. To everyone's surprise, Harry Potter is chosen to participate in the competition, in which he must fight dragons, enter the maze and face his greatest fears.	2000-07-08	J.K. Rowling (J.K. Rowling)	2024-05-29	2024-05-29	[Edit] [Delete]
7	The Hobbit's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comical misadventures of Arthur Dent, a hapless Englishman following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens. Arthur escapes with his friend Ford Prefect, who reveals himself to be an alien and a member of the Galactic Patrol. The series follows the adventures of the two men as they travel through space, encountering various alien species and the dangers of the universe.	1978-10-17	Douglas Adams	2024-05-29	2024-05-29	[Edit] [Delete]
8	Frankenstein; or the Modern Prometheus	A Gothic novel by Mary Shelley that tells the story of Victor Frankenstein, a young scientist who creates a grotesque creature in an unorthodox scientific experiment. Frankenstein is terrified by his creation and abandons it, but the creature seeks revenge. The novel explores themes of ambition, responsibility, creation versus destruction, and the nature of good and evil.	1818-03-01	Mary Shelley	2024-05-29	2024-05-29	[Edit] [Delete]
9	The Lord of the Rings: The Fellowship of the Ring	The first book in J.R.R. Tolkien's epic fantasy trilogy, The Lord of the Rings. The Fellowship of the Ring follows hobbit Frodo Baggins as he inherits the One Ring, an evil artifact of power created by the Dark Lord Sauron. Frodo embarks on a quest to destroy the Ring in the fires of Mount Doom, accompanied by a fellowship of eight companions.	1954-07-29	J.R.R. Tolkien	2024-05-29	2024-05-29	[Edit] [Delete]
10	Shogun	This is only my testing purpose.	2024-10-11	Samuel Fuller	2024-05-29	2024-05-29	[Edit] [Delete]

6.3 Books Management Page

6. Application User Interface

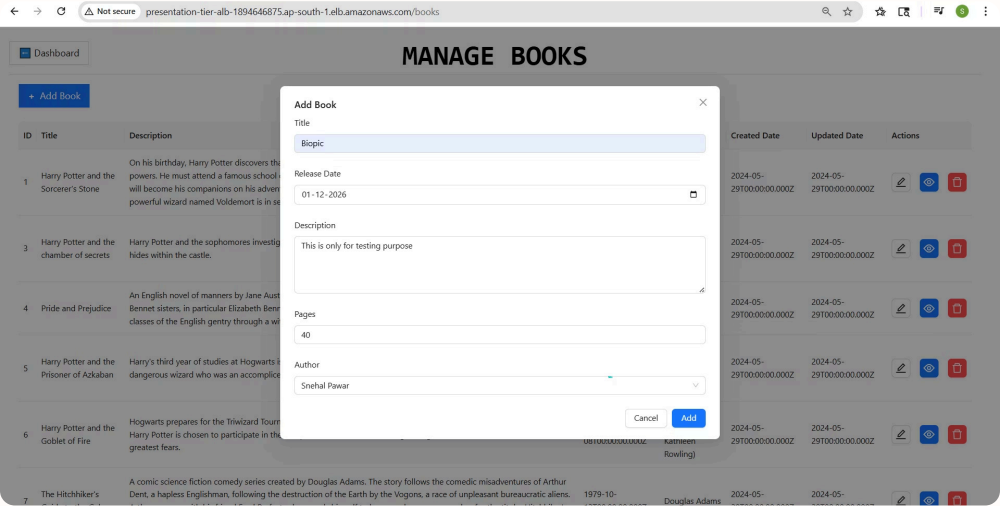


6.3 Adding a New Author



6.4 Author Data Successfully Added

6. Application User Interface

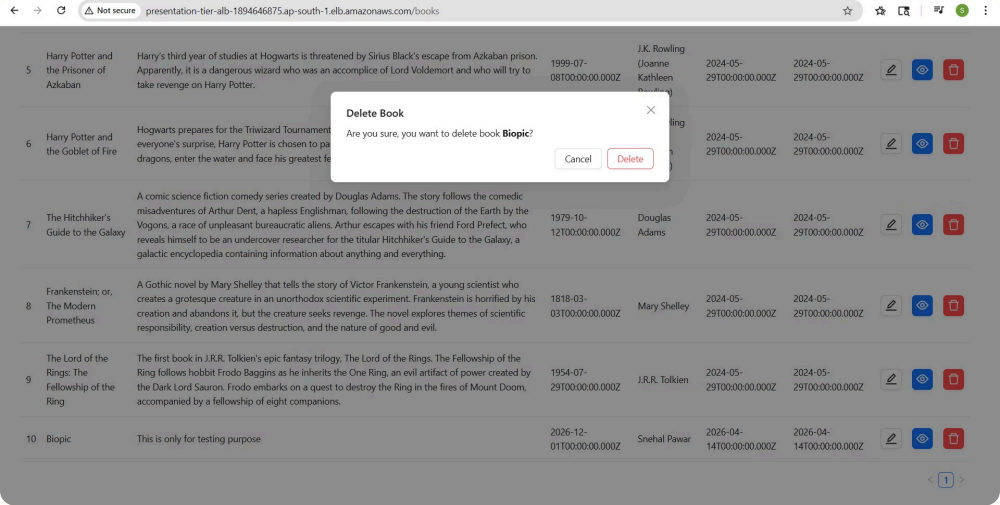


6.3 Adding a New Book

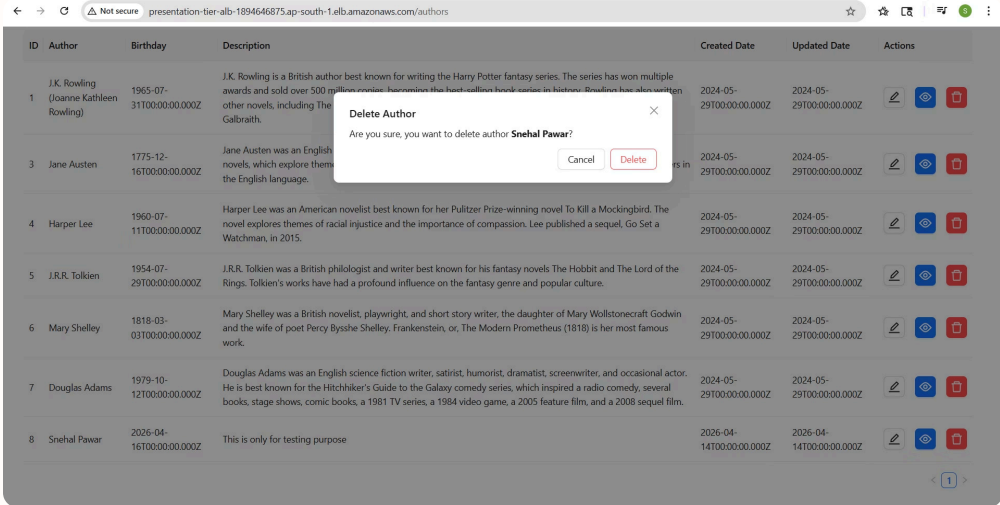
ID	Title	Description	Release Date	Author	Created Date	Updated Date	Actions
1	Harry Potter and the Sorcerer's Stone	On his birthday, Harry Potter discovers that he is the son of two well-known wizards, from whom he has inherited magical powers. He must attend a famous school of magic and sorcery, where he establishes a friendship with two young men who will become his companions on his adventure. During his first year at Hogwarts, he discovers that a malevolent and powerful wizard named Voldemort is in search of a philosopher's stone that prolongs the life of its owner.	1997-07-26T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
3	Harry Potter and the chamber of secrets	Harry Potter and the sophomores investigate a malevolent threat to their Hogwarts classmates, a menacing beast that hides within the castle.	1998-07-02T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
4	Pride and Prejudice	An English novel of manners by Jane Austen, first published in 1813. The story centres on the relationships among the Bennet sisters, in particular Elizabeth Bennet the middle daughter, and the wealthy Mr. Darcy. Austen satirizes the social classes of the English gentry through a witty and ironic narrative voice.	1813-01-28T00:00:00.000Z	Jane Austen	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament, in which three schools of wizardry will compete. To everyone's surprise, Harry Potter is chosen to participate in the competition, in which he must fight dragons, enter the water and face his greatest fears.	2000-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
7	The Hitchhiker's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comedic misadventures of Arthur Dent, a hapless Englishman, following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens. Arthur escapes with his friend Ford Prefect, who reveals himself to be an undercover researcher for the titular Hitchhiker's Guide to the Galaxy, a galactic encyclopedia containing information about anything and everything.	1979-10-12T00:00:00.000Z	Douglas Adams	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
8	Frankenstein; or, The Modern Prometheus	A Gothic novel by Mary Shelley that tells the story of Victor Frankenstein, a young scientist who creates a grotesque creature in an unorthodox scientific experiment. Frankenstein is horrified by his creation and abandons it, but the creature seeks revenge. The novel explores themes of scientific responsibility, creation versus destruction, and the nature of good and evil.	1818-03-03T00:00:00.000Z	Mary Shelley	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
9	The Lord of the Rings: The Fellowship of the Ring	The first book in J.R.R. Tolkien's epic fantasy trilogy, The Lord of the Rings. The Fellowship of the Ring follows hobbit Frodo Baggins as he inherits the One Ring, an evil artifact of power created by the Dark Lord Sauron. Frodo embarks on a quest to destroy the Ring in the fires of Mount Doom, accompanied by a fellowship of eight companions.	1954-07-29T00:00:00.000Z	J.R.R. Tolkien	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
10	Biopic	This is only for testing purpose	2026-12-01T00:00:00.000Z	Snehal Pawar	2026-04-14T00:00:00.000Z	2026-04-14T00:00:00.000Z	[Edit] [Share] [Delete]

6.4 Book Data Successfully Added

6. Application User Interface

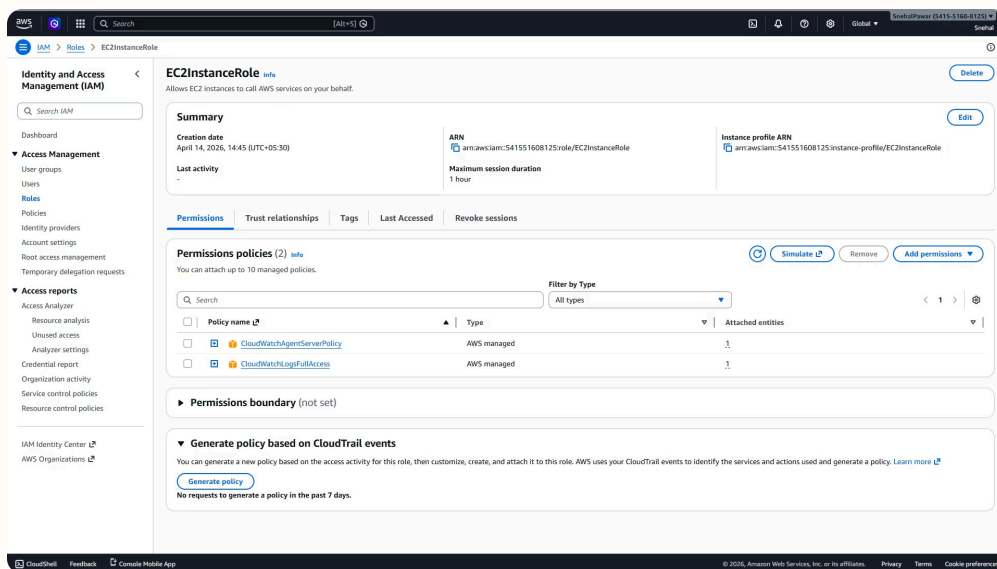


6.5 Deleting a Book Record

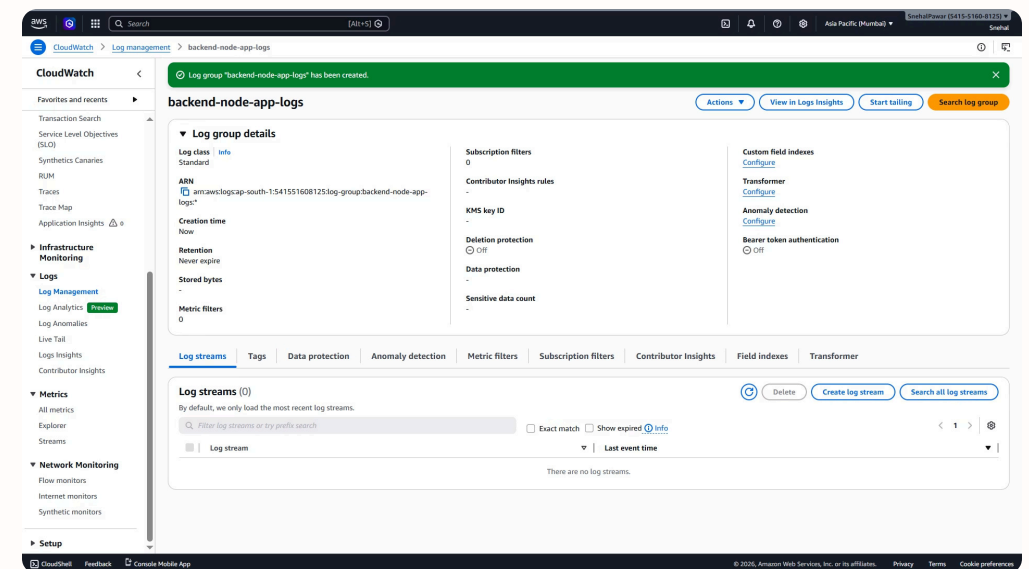


6.6 Deleting an Author Record

7. Monitoring and Logging



7.1 IAM Role Creation



7.2 CloudWatch Log Group Setup

7. Monitoring and Logging

The screenshot displays the AWS CloudWatch Logs console interface. The top navigation bar includes the AWS logo, a search bar, and user information (SnehalPawar, 5415-5160-8125). The breadcrumb trail shows the path: CloudWatch > Log management > backend-node-app-logs > i-070ecf352b1adebbd.

The left sidebar contains the following sections:

- CloudWatch**
 - Favorites and recents
 - Application Signals (APM)**
 - Services
 - Application Map
 - Transaction Search
 - Service Level Objectives (SLO)
 - Synthetics Canaries
 - RUM
 - Traces
 - Trace Map
 - Application Insights
 - Infrastructure Monitoring**
 - Logs**
 - Log Management**
 - Log Analytics
 - Log Anomalies
 - Live Tail
 - Logs Insights
 - Contributor Insights
 - Metrics**

The main content area is titled **Log events** and includes a filter bar with the text "Filter events - press enter to search". Below the filter bar, there are buttons for "Clear", "1m", "30m", "1h", "12h", "Custom", "UTC timezone", and "Display".

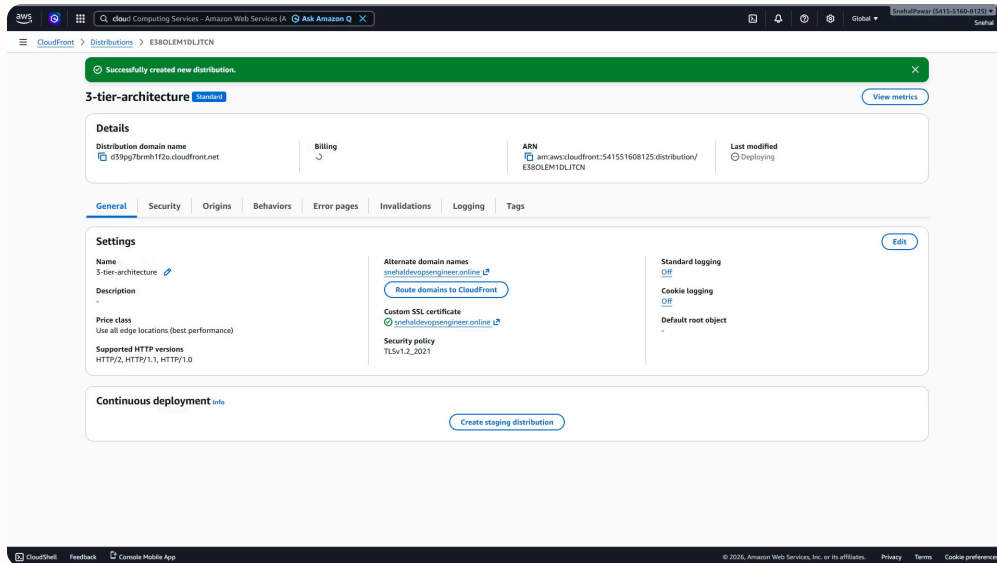
The log events are displayed in a table with two columns: **Timestamp** and **Message**.

Timestamp	Message
No older events at this moment. Retry	
2026-04-14T10:40:14.000Z	2026-04-14 10:40:14 [INFO]: Connected to MySQL Database
2026-04-14T10:42:48.000Z	2026-04-14 10:42:48 [INFO]: BooksController [CREATE] - title: Biopic, description: This is for Testing Purpose, releaseDate: 2026-12-01, pages: 50, authorId: 9
2026-04-14T10:42:48.000Z	2026-04-14 10:42:48 [INFO]: Book created successfully. books count: 9
2026-04-14T10:42:54.000Z	2026-04-14 10:42:54 [INFO]: BooksController [GET]
2026-04-14T10:42:54.000Z	2026-04-14 10:42:54 [INFO]: Books count: 9
2026-04-14T10:45:06.000Z	2026-04-14 10:45:06 [INFO]: AuthorsController [GET]
2026-04-14T10:45:06.000Z	2026-04-14 10:45:06 [INFO]: Authors count: 7
2026-04-14T10:45:15.000Z	2026-04-14 10:45:15 [INFO]: BooksController [DELETE] - bookId: 11
2026-04-14T10:45:15.000Z	2026-04-14 10:45:15 [INFO]: Book deleted successfully. books count: 8
No newer events at this moment. Auto retry paused . Resume	

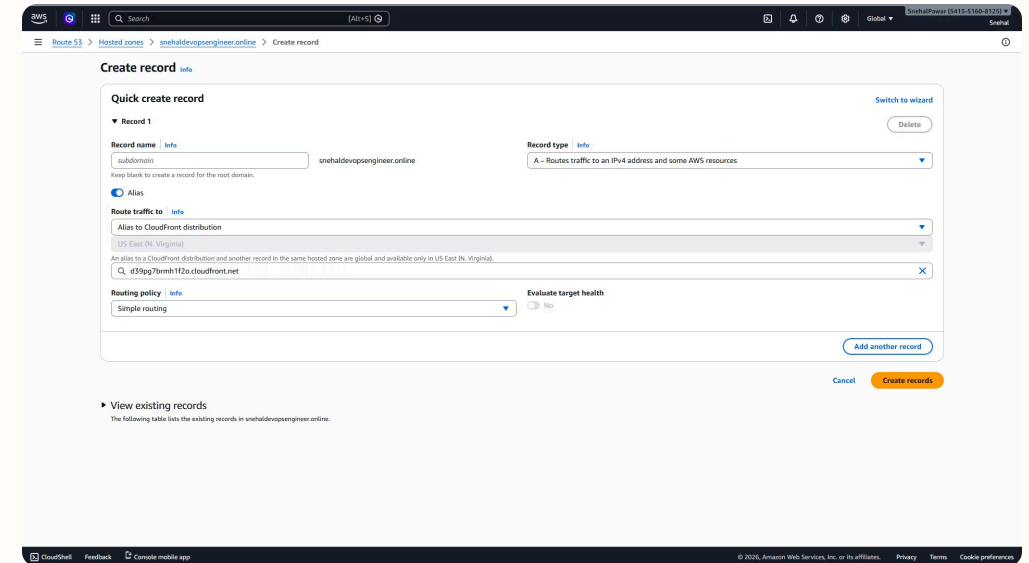
The bottom of the console shows links for CloudShell, Feedback, and Console Mobile App, along with the copyright notice: © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences.

7.3 CloudWatch Logs

8. Content Delivery and DNS Mapping

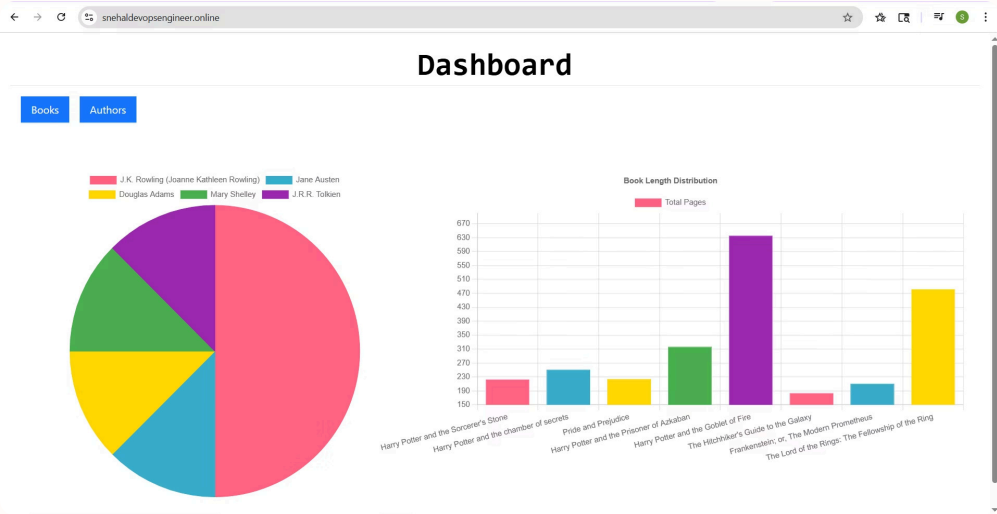


8.1 CloudFront Distribution Setup

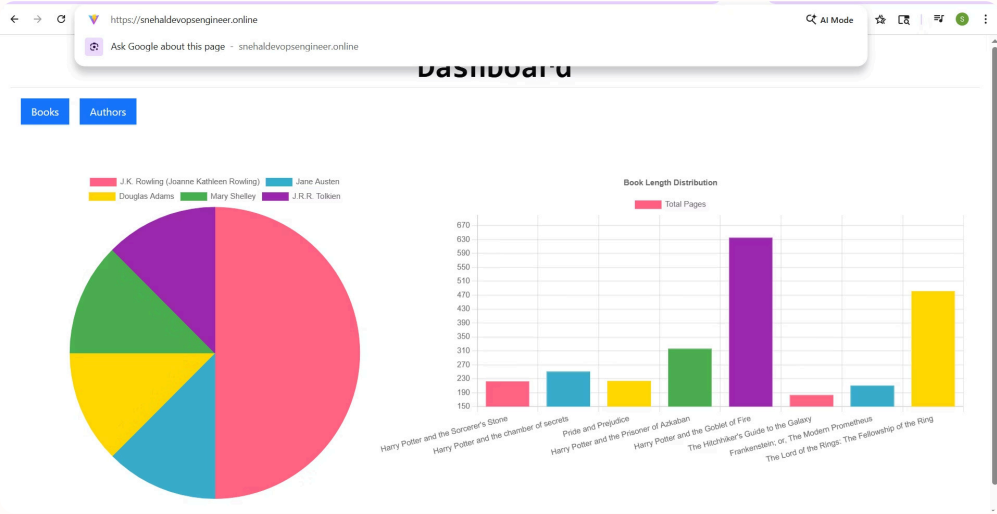


8.2 DNS Alias Record Configuration for CloudFront

9. Final Output



9.1 Application Access via Application Load Balancer DNS



9.2 Application Access via Custom Domain



Key Achievements



Successfully deployed 3-tier architecture



Implemented Auto Scaling



Integrated CloudWatch logging



Configured CloudFront for performance



Achieved high availability



Future Scope



Add CI/CD pipeline

Jenkins / GitHub Actions



Use Docker & Kubernetes



Implement advanced security

Web Application Firewall (WAF)



Add caching

Redis



Improve monitoring dashboards

Project Resources



GitHub Repository



AWS Documentation



YouTube Tutorials



Internship Materials



Learning from Internship



Hands-on AWS experience



Understanding of cloud architecture



Deployment of real-world application



Debugging & monitoring skills



DevOps fundamentals

Thank You! 🙌

Thank you for your attention!

Let's Connect

Questions, feedback, or discussion are always welcome.

Presenter

Snehal B. Pawar

Roll No. 5649

S. M. Joshi College, Pune

Questions?

